

FAXMUSIC: Social Network for Musicians

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I - INTRODUCTION:

Project Name: faxmusic

Objective: Create a social platform allowing musicians to share their music, collaborate and interact with their audience.
Target audience: Amateur and professional musicians, producers, music fans.

II · ANALYSIS:

Analysis of existing systems:

1 - Existence of competing platforms

- SoundCloud: Platform allowing the sharing and listening of music tracks, but with few direct collaboration tools.
- Bandcamp: Allows musicians to sell their music directly to fans, but does not offer advanced social features.
- Facebook & Instagram: Used by artists for promotion, but are not specifically designed for musicians.
- ReverbNation: Offering music promotion tools, but with a limited community.

2 - Opportunities for faxmusic

- Create a network specifically tailored to the needs of musicians.
- Integrate novel collaborative tools (audio file sharing, real-time discussion, musical workspace).
- Offer direct and simplified monetization solutions for artists.

Needs Analysis:

1 - Functional Requirements:

- User Management:

- login
- registration
- music profile management

- Content Management:

- Uploading of audio, video, and image files
- Publishing posts (text, images, audio/video files)
- Integration of audio and video players
- Comments and likes on posts

- Interaction and collaboration:

- Private messaging system between users
- Forums and groups by musical style
- Marketplace for selling beats, musical services, instruments

- Event Management
 - Creation and sharing of musical events (concerts, festivals)

2 - Non-functional Requirements:

- Accessibility: Interface adapted to accessibility standards
- Ergonomics: Intuitive user experience with a responsive design.
- Performance: Optimized loading times, content caching.
- Reliability: High availability with regular data backup.
- Security: Data encryption, protection against attacks
- Compatibility: Operation on desktop, tablet, and mobile and Web application
- Scalability: Ability to scale the infrastructure based on user growth.

2 - Technical Requirements:

Technology to use:

- Frontend: React (Vite) with Tailwind CSS
- Mobile: React Native
- Backend: Express.js with MySQL
- File storage: Firebase Storage
- WebSockets: socket.IO (node js) for notifications and messaging

III · SPECIFICATIONS:

FUNCTIONAL UNITS:

Module: User Management

N°UF01	login
Description	Log in to be able to access the content of the application or site
constraints	Have an account on the platform
Input	Username (username minimum 4 characters), password (password minimum 8 characters)
Output	Loading and redirection to the home page if the username and password are correct, in case of error or incorrect information display of error message or incorrect info, and in case of missing information
user	New amateur or professional musicians

N°UF02	Registration
Description	Registration for new users
constraints	Have an email address
Input	Username (username minimum 4 characters), password (password minimum 8 characters),

	email address, last name and first name
Output	Loading and redirection to the music profile management page, and in case of error display error, and in case of already existing username display message, and in case of missing information
user	User who has an account

N°UF03 Music profile management	
Description	Allows adding a profile picture, musical genre, amateur or professional status, and instrument
constraints	Knowing one's musical genre and having a profile picture
Input	Musical genre, selection of instrument type, professional or amateur
Output	Loading and redirection to the login page, and in case of error display error, and in case of missing information
user	User who has already registered

Module : Content Management

N°UF01 Uploading audio, video, and image files	
Description	Allows uploading files
constraints	Must be logged in and have an account
Input	Upload button
Output	Display of upload progress
user	Logged-in user

N°UF02	Publishing posts (text, images, audio/video files)
Description	Publish files and texts publicly
constraints	Must be logged in and have an account
Input	Publish button
Output	Display of file selection or text area
user	Logged-in user

N°UF03	Integration of audio and video players
Description	Allowing to play the published audio and video content
constraints	Must be logged in and have an account
Input	Press on the content to play
Output	Display of the audio or video player
user	Logged-in user

N°UF04	Comments and likes on posts
Description	Allowing to play the published audio and video content
constraints	Must be logged in and have an account
Input	Press on the bottom of the content
Output	Display of the text area for the comment and display of the reaction for likes

user	Logged-in user
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Module : Interaction and collaboration

N°UF01 Private messaging system between users	
Description	Allowing communication between musicians
constraints	Must be logged in and have an account
Input	Press on the messaging menu
Output	Display of the messaging interface
user	Logged-in user

N°UF02 Forums and groups by musical style	
Description	Allowing communication between musicians of the same music genre
constraints	Must be logged in and have an account
Input	Press the join a group button
Output	Display of the interface
user	Logged-in user

N°UF03 Marketplace for selling beats, musical performances, instruments	
Description	Allowing communication between musicians

constraints	Must be logged in and have an account
Input	Press on the music store creation menu
Output	Display of the creation interface
user	Logged-in user

Module : Event Management

N°UF01	Creation and sharing of musical events (concerts, festivals)
Description	Allowing to publish musical events
constraints	Must be logged in and have an account
Input	Press on the event creation menu
Output	Display of the event creation interface
user	Logged-in user